Code No: 155BT **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, August - 2022
IMAGE PROCESSING
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) What is digital image? Explain the fundamental steps in digital image processing.
b) Explain the Concept of Gray Levels and its conversion to Binary Image. [8+7]

2. Explain in detail about DFT, DCT, KLT and SVD techniques. [15]

3. Differentiate and explain the concept of Image Enhancement in Spatial Domain and Frequency Domain. [15]

4.a) Explain histogram processing.
b) Distinguish between image smoothing and sharpening. [7+8]

5.a) Discuss about the Image Restoration Degradation Model with a neat diagram.
b) Explain how Image Restoration can be done using various filters? [7+8]

6. Explain how image degradation is estimated using observation and mathematical modelling. [15]

7.a) Explain basic global thresholding with iterative algorithm.
b) Discuss the various techniques used in image segmentation. [7+8]

8.a) Explain the image compression model with a neat diagram.
b) Differentiate and explain the various compression techniques. [7+8]

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